

report01.2_data_science_MahdiGhadmi

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Customer Personality Analysis

1 Introduction

The goal of this exercise is to analyze a [customer personality dataset](#) containing demographic information, spending habits, promotional responses, and purchasing behaviors. We will conduct hypothesis tests to uncover insights about customer behaviors and characteristics:

- Do customers from different education levels have different income levels?
- Is there a significant difference in spending before and after a marketing campaign?
- Do customers with children spend differently than those without children?
- Is there a significant difference in customer satisfaction across multiple product categories?
- Did the number of purchases change significantly after a new marketing strategy?

2 Data Exploration & Preprocessing

2.1 Dataset Overview

- The dataset consists of **2240 rows** and **29 columns**.

Column	Non-Null Count	Dtype
ID	2240 non-null	int64
Year_Birth	2240 non-null	int64
Education	2240 non-null	object
Marital_Status	2240 non-null	object
Income	2216 non-null	float64
Kidhome	2240 non-null	int64
Teenhome	2240 non-null	int64
Dt_Customer	2240 non-null	object
Recency	2240 non-null	int64
MntWines	2240 non-null	int64
MntFruits	2240 non-null	int64
MntMeatProducts	2240 non-null	int64
MntFishProducts	2240 non-null	int64
MntSweetProducts	2240 non-null	int64
MntGoldProds	2240 non-null	int64
NumDealsPurchases	2240 non-null	int64

Column	Non-Null Count	Dtype
NumWebPurchases	2240 non-null	int64
NumCatalogPurchases	2240 non-null	int64
NumStorePurchases	2240 non-null	int64
NumWebVisitsMonth	2240 non-null	int64
AcceptedCmp3	2240 non-null	int64
AcceptedCmp4	2240 non-null	int64
AcceptedCmp5	2240 non-null	int64
AcceptedCmp1	2240 non-null	int64
AcceptedCmp2	2240 non-null	int64
Complain	2240 non-null	int64
Z_CostContact	2240 non-null	int64
Z_Revenue	2240 non-null	int64
Response	2240 non-null	int64

2.2 Handling NaN Data and Data Cleaning

- The “Income” column had 24 null values that were dropped

3 Statistical Analysis

3.1 Do customers from different education levels have different income levels?

- Method: Kruskal-Wallis H test
- Analysis:
 - The Kruskal-Wallis H test was used to compare the income levels of customers from different education levels (UG, Master, PhD, Graduation).
 - Kruskal-Wallis H-statistic: 71.99833577587502
 - P-value: 1.5932077492910976e-15
- Result:

The p-value (1.59e-15) indicates a highly significant difference in income levels across education levels.

3.2 Is there a significant difference in spending before and after a marketing campaign?

- Method: Mann-Whitney U Test
- Analysis:
 - The Mann-Whitney U test compared the total spending (Total_spents) of customers who accepted at least one campaign vs. those who did not.
 - TU statistic: 255968.0
 - P-value: 1.300457942953344e-66
- Result:

The p-value (1.30e-66) shows a very significant difference in spending before and after a marketing campaign.

3.3 Do customers with children spend differently than those without children?

- Method: Mann-Whitney U Test
- Analysis:
 - The Mann-Whitney U test was applied to compare the spending of customers with children ($\text{Total_children} > 0$) and those without children ($\text{Total_children} = 0$).
 - Mann-Whitney U-statistic: 798765.0
 - P-value: 3.7522691035343634e-106

- Result:

The p-value (3.75e-106) indicates a very significant difference in spending.

3.4 Is there a significant difference in customer satisfaction across multiple product categories?

- Method: Friedman Test
- Analysis:
 - The Friedman test was used to assess satisfaction across different product categories (wine, fruits, meat products, fish products, sweets, and gold).
 - Friedman Test Statistic: 5889.561490535339
 - P-value: 0.0

- Result:

The p-value (0.0) shows a very significant difference in satisfaction across product categories.

3.5 Did the number of purchases change significantly after a new marketing strategy?

- Method: chi2_contingency
- Analysis:
 - A chi-square test was used to evaluate the relationship between campaign participation ($\text{Total_AcceptedCmps}$) and education level.
 - Chi-Square Statistic: 22.507501142376434
 - P-Value: 0.09517110155293661

- Result:

The p-value (0.095) indicates no significant difference between the number of purchases and the education level of customers.

4 Discussion & Conclusion

Our analysis focused on understanding the spending behavior and preferences of customers based on various characteristics, such as education level, campaign participation, children in the household, and product categories. We applied statistical tests to evaluate whether significant differences existed across these factors.

1. We suggest that customers' income varies significantly depending on their education level. Marketing strategies could tailor income-specific offers or promotions. For example, higher-income customers (often with more advanced education) may be more inclined to purchase premium products or services.
2. Customers who engage with campaigns tend to spend significantly more than those who do not. This demonstrates the effectiveness of marketing campaigns. Marketing teams could use this insight to increase campaign participation, perhaps by offering more compelling promotions to encourage engagement.
3. Parents tend to spend more than non-parents. This could suggest that families have different purchasing patterns, possibly due to increased needs for certain products (e.g., food, clothing, or entertainment). Targeted marketing strategies, such as family-oriented promotions, could lead to increased sales from this group.
4. Customers are more satisfied with some product categories than others. This suggests that businesses could focus on improving the products or services that receive lower satisfaction ratings, ensuring customers have a more consistent experience across categories. Additionally, products that already have high satisfaction might be highlighted more in marketing efforts.
5. The marketing strategy did not significantly affect purchase behavior based on education. This might suggest that the new marketing strategy does not cater specifically to education-based customer segments or that other factors (such as campaign content or timing) play a more significant role in driving purchases.

5 Appendix

5.1 Business Actionable Insights(as Data Storytelling)

1. Segmented Marketing Based on Education: Since income varies significantly by education level, tailor marketing strategies to specific education groups. For instance, higher-income groups might be offered premium products, while lower-income groups could receive discounts or value-based promotions.
2. Maximizing Campaign Effectiveness: The significant difference in spending before and after a campaign indicates that marketing campaigns are effective in increasing spending. To enhance future campaigns, marketers could analyze which campaigns led to the highest spending and replicate those strategies. Offering personalized, targeted discounts or promotions might increase engagement and customer spending.
3. Targeting Families: Parents tend to spend more, which means family-oriented products or packages could be a profitable area to focus on. This group might respond positively to family-centered offers, such as bundled deals or loyalty programs tailored to family needs.

4. Improving Product Categories with Lower Satisfaction: Products that scored poorly in satisfaction (e.g., certain categories like sweets or fish products) should be improved to enhance customer experience. Conversely, popular product categories could be featured more prominently in advertising and sales promotions.
5. Review and Refine Marketing Strategies: The lack of significant change in purchases after the marketing strategy suggests that other factors, like product type or timing, need to be better aligned with the target audience's preferences. Future campaigns could test different strategies, such as personalized messages, time-sensitive promotions, or more frequent communication with customers.